

Highlights

Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

Haoran Ma; Ningning Wang

- SC-FMA converts knowledge-intensive traces into auditable knowledge-priority records.
- The method combines annotation fidelity, structural dependency, redundancy, and bottleneck roles.
- PRM800K evidence supports fixed-budget audit prioritization under annotation signals.
- Knowledge-graph representation checks show structural-dependency allocation sensitivity.
- Budget-constrained audit-target coverage improves from 0.235 to 0.699 over a scalar-only view.